

The Polak–Schrijver code has exactly eight private pairs

Artem Oktiabrev*

Independent researcher, Ukraine

Abstract

Every improvement on the lower bound for the Shannon capacity $\Theta(C_7)$ since July 2026 is built on one five-dimensional gadget: the 367-word independent set of Polak and Schrijver together with eight *private pairs* — vertices outside the code with a single neighbour in it — and the private-pair count is the parameter those recursions are most sensitive to. We prove that eight is the exact maximum: in the whole of $\mathbb{Z}_7^5$ exactly eight vertices have a single neighbour in the code, they are precisely the eight in use, and all eight are simultaneously usable because their confusability graph is a matching, hence bipartite. A ninth private pair for this code does not exist. We also report the exhaustive computations around the theorem: all eight codes the Polak–Schrijver pipeline can produce, with counts 5, 6, 6, 6, 7, 7, 7, 8; the arithmetic that rules out smaller codes; and a sweep of 8,260,976,640 images under $\text{Aut}(C_7^{\boxtimes 5})$ which shows why the published construction has to replace one auxiliary vector.

Keywords: Shannon capacity, zero-error capacity, independent set, strong product, odd cycle, gadget product

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1. Context, and what the question is worth

For a graph G , the Shannon capacity is $\Theta(G) = \sup_d \alpha(G^{\boxtimes d})^{1/d}$ [1]. For the seven-cycle it is unknown, and the gap between the best bounds is wide: the Lovász bound gives $\Theta(C_7) \leq \vartheta(C_7) = 3.3176672073940953927\dots$ [2], while the lower bound has moved only in the fourth decimal place in seven years. Table 1 lists the chain.

The structure behind the last four rows is Gao’s gadget [7], restated as a *valid tuple* in [8] and refined in [9]: an independent set I (the code), t private pairs, two complementary transversals of their endpoints, and an auxiliary independent set X . The product lemma multiplies gadgets, and the parameter sensitivity of the resulting bound is lopsided. Evaluated at hypothetical base profiles (a, t, s, o) , the homogeneous recursion of [7, 8] at its optimal

date	$\Theta(C_7) \geq$	from	with
1971–2015	$343^{1/5}$, then $350^{1/5}$	[3, 4]	a li
2019	3.2578659667835159504124...	[5]	$\alpha(C_7)$
Jul 2026	3.2580207372932453595278...	[6]	$\alpha(C_7)$
Jul 2026	3.2587891539086910161967650155206769...	[7]	rec
Jul 2026	3.2588053698854655725829750306238067...	[8]	a st
Aug 2026	3.2588326203532663091215390518104754...	[9]	het

Table 1: Lower bounds on $\Theta(C_7)$. The last four entries all start from the same 367-word code and the same eight private pairs; they differ in the recursion built on top.

depth gives

$$(367, 8, 367, 322) \mapsto 3.2588053698854655725829750306238067\dots,$$

so one further private pair is worth $1.736 \cdot 10^{-4}$ over the current record, whereas one further auxiliary word of the neutral class is worth about a tenth of that ($(367, 8, 367, 324)$, two extra such words, clears the record by only $6.993 \cdot 10^{-6}$). For scale: the difference between the record [9] and what the homogeneous recursion extracts from the *same* base gadget is $2.725 \cdot 10^{-5}$, so a ninth private pair would improve the state of the art under any of the pub-

*Corresponding author.

Email address: aoktyabrev@gmail.com (Artem Oktiabrev)

URL: <https://orcid.org/0009-0003-3626-2002> (Artem Oktiabrev)

lished recursions, by roughly seven times the spread between the published schemes. That is why the question of a ninth pair is worth settling rather than leaving as folklore, and it is the question this note answers.

The answer is negative and exact, which also explains a pattern in the literature: the three independent lines of work after [6] all keep the same base gadget and improve the recursion around it. On that code the private-pair count cannot be raised at all.

2. Gadgets and private pairs

Vertices of $C_7^{\boxtimes d}$ are the elements of $\mathbb{Z}_7^d$; two distinct vertices are adjacent (*confusable*) iff their circular distance is at most 1 in every coordinate. We write $N[v]$ for the closed neighbourhood and call $I \subseteq \mathbb{Z}_7^d$ independent if no two of its elements are confusable.

Definition 1 (private pair, [7]). Let I be independent. A pair (r, q) with $r \in I$, $q \notin I$ is a *private pair* for I if $N[q] \cap I = \{r\}$. The vertex r is the *centre* and q the *private neighbour*.

Definition 2 (gadget, [7]). A gadget consists of an independent set I of size a ; private pairs (r_i, q_i) , $1 \leq i \leq t$, with pairwise disjoint endpoint sets; two complementary transversals P^H, P^V of the endpoints — for every i exactly one of r_i, q_i lies in P^H and the other in P^V — both independent; and an independent auxiliary set X with $X \cap N(P^H) \cap N(P^V) = \emptyset$. With $O = X \setminus (N(P^H) \cup N(P^V))$, $H = X \cap N(P^H)$, $V = X \cap N(P^V)$, the profile is $(a, t, s, o, h, v) = (a, t, |X|, |O|, |H|, |V|)$.

Write $t^*(I)$ for the largest t for which some gadget with code I exists. The published base gadget has $I = I_0$, the 367-word set printed in the appendix of [5], and $t = 8$ with the eight pairs of [6]; its profile is $(367, 8, 367, 321, 26, 20)$ in [7] and $(367, 8, 367, 322, 26, 19)$ in [8, 9].

3. The theorem

The definition of a gadget quantifies over transversals, which is what makes t^* look hard to bound. It is not: the transversal condition collapses to a condition on the private neighbours alone.

Lemma 3 (no mixing). Let (r_i, q_i) and (r_j, q_j) be private pairs for I with $r_i \neq r_j$. Then r_i is confusable with neither r_j nor q_j .

Proof. $r_i, r_j \in I$ and I is independent, so $r_i \not\sim r_j$. If $r_i \sim q_j$ then $r_i \in N[q_j] \cap I = \{r_j\}$, so $r_i = r_j$, contradicting the hypothesis. $\square$

Corollary 4. Let (r_i, q_i) , $1 \leq i \leq t$, be private pairs for I with pairwise distinct centres, and let $\sigma : \{1, \dots, t\} \rightarrow \{H, V\}$ assign to each pair the transversal holding q_i . Then the endpoint sets are pairwise disjoint, and P^H, P^V are both independent if and only if σ is a proper 2-colouring of the graph Γ on $\{q_1, \dots, q_t\}$ whose edges are the confusable pairs $q_i \sim q_j$. Consequently $t^*(I)$ is the largest number of private pairs with distinct centres whose q -graph Γ is bipartite.

Proof. Distinct centres give $r_i \neq r_j$; $q_i \neq q_j$ because $N[q_i] \cap I = \{r_i\}$ pins the centre; and $r_i \neq q_j$ because $r_i \in I$, $q_j \notin I$. So the endpoint sets are disjoint. Each transversal consists of some r 's and some q 's. Two r 's are never confusable, and an r and a q of different pairs are never confusable by Lemma 3; an r_i and q_i of the *same* pair are confusable but lie in different transversals by construction. Hence a transversal can fail to be independent only by containing two confusable private neighbours, which is exactly a monochromatic edge of Γ . $\square$

So the only candidates that need to be examined are the private pairs of I themselves, and they form a finite list which can be written down: every $q \notin I$ with $|N[q] \cap I| = 1$, together with that single neighbour. For the 367-word code this list is short.

Proposition 5 (the enumeration). Let $I_0 \subseteq \mathbb{Z}_7^5$ be the 367-word Polak–Schrijver code as printed in [5]. Of the $16807 - 367 = 16440$ vertices outside I_0 , exactly 8 have one neighbour in I_0 , 254 have two, and 16178 have three or more; none has zero. The eight are exactly the private neighbours $q_0, \dots, q_7$ of [6, 7], with the centres those papers give them.

Proof. One pass over $\mathbb{Z}_7^5$, counting for each vertex outside I_0 its neighbours in I_0 ; the predicate is the integer condition of Section 2, so the computation is exact. The empty zero-bucket says that I_0 is a maximal independent set, which is independently known [5, Appendix]. The identification with the published pairs is a set comparison. See Section 6. $\square$

Theorem 6. $t^*(I_0) = 8$. Moreover the admissible transversal assignments are exactly the proper 2-colourings of a graph with five connected components, i.e. 16 of them up to exchanging P^H and

P^V , and one of these is the assignment used in [6, 7, 8, 9].

Proof. By Corollary 4, $t^*(I_0)$ is computed from the candidate list of Proposition 5. That list has eight entries with eight distinct centres, so $t^*(I_0) \leq 8$. Its conflict graph Γ has, in the numbering of [7], exactly the three edges

$$q_0 \sim q_1, \quad q_2 \sim q_6, \quad q_3 \sim q_5,$$

i.e. Γ is a perfect matching on six of the eight vertices and has two isolated vertices. A matching is bipartite, so all eight pairs may be used at once and $t^*(I_0) = 8$. The number of components of Γ is $3 + 2 = 5$, and a graph whose components are single edges and isolated vertices has 2^5 proper 2-colourings, i.e. $2^4 = 16$ up to a global swap; that the published assignment ($J_0 = \{0, 5, 6\}$ carrying r_j into P^H , $J_1 = \{1, 2, 3, 4, 7\}$ the other way, as in [7]) is one of them is checked directly. $\square$

Corollary 7. *No gadget whose code is I_0 has $t \geq 9$. In particular the profile $(367, 9, 367, \cdot)$, which would give $\Theta(C_7) \geq 3.2590062007738521305802 \dots$ through the homogeneous recursion of [7, 8], is not reachable from the Polak–Schrijver code, and the same holds for every recursion that takes a gadget over this code as its base, including [9].*

Lemma 3 and Corollary 4 are proved by hand; the two finite facts they are applied to — the candidate list and the three edges — are direct computations over 16807 vertices, small enough for any reader to repeat from the printed set.

4. The exhaustive computations around the theorem

Theorem 6 is about one code. Three further questions are settled by computation rather than by proof, and together they close the directions in which a ninth pair could have been sought.

All codes the Polak–Schrijver construction can produce. Step (v) of [5] extends a core of 327 words by a maximum independent set of an extension graph with 71 vertices and 85 edges, whose independence number is 40. That maximum is not unique. Enumerating all of them gives exactly 8 distinct codes of size 367; their private-pair counts, each computed as in Theorem 6, are

$$5, 6, 6, 6, 7, 7, 7, 8,$$

and the unique code attaining 8 is the printed one. The construction that produced the record therefore has nothing better to give: the literature is already using the best member of the family.

Codes of other sizes. A gadget’s code need not be maximum, so a smaller code with more private pairs is admissible in principle. The published recursion prices that trade exactly. Matching the bound of $(367, 8, 367, 321)$ would require

$$13 \text{ private pairs at } a = 366, \quad 16 \text{ at } a = 365, \quad 23 \text{ at } a = 360,$$

while the 367-word code has eight candidates in the whole of $\mathbb{Z}_7^5$. One lost code word costs about six times what an extra private pair gains, so this branch is closed by arithmetic and not by search.

The auxiliary set, over the whole automorphism group. With $t = 8$ forced, the remaining parameter is X . Since $|X|$ should stay at 367, the candidates are the eight pipeline codes and their images under $\text{Aut}(C_7^{\boxtimes 5}) \supseteq (D_7)^5 \rtimes S_5$, a group of $14^5 \cdot 120 = 64,538,880$ elements; every admissible 2-colouring changes $N(P^H)$ and $N(P^V)$, so the sweep covers $8 \cdot 16 \cdot 64,538,880 = 8,260,976,640$ triples (source code, colouring, automorphism). It is exhaustive, not sampled: for fixed permutation and reflections all 16807 translations are scored at once by a cyclic cross-correlation. The result is that the forbidden intersection $X \cap N(P^H) \cap N(P^V)$ is never empty — the minimum over all 8,260,976,640 triples is one word:

words of X in the forbidden region	0	1	2	3	4	5
triples	0	4	33	101	209	358

This explains a step of the published construction that is stated there without comment: Gao takes X to be the image $T(I_0)$ under $T(w) = (2 - w_1, w_3, w_0, 2 - w_2, w_4)$ with the single word $(2, 4, 6, 3, 5)$ replaced by $(1, 5, 6, 3, 5)$ [7, 9]. The replacement is unavoidable rather than cosmetic, and $(2, 4, 6, 3, 5)$ is exactly the one word of $T(I_0)$ confusable with both transversals. Repairing the near misses exactly — delete the offending words, then solve for the best legal replacement — reaches at best $s = 367$, $o = 322$: the Buys–Polak–Zuiddam profile [8], recovered from another direction and not improved. Across the 60 repaired candidates no repair admitted more insertions than deletions; had

one, it would have been an independent set of size 368 in $C_7^{\boxtimes 5}$, which the script watched for and did not see.

5. Two observations, with their status

Observation 8 (the colouring the literature uses may be the only usable one). Of the 16 proper 2-colourings of Theorem 6, a 367-word admissible auxiliary set was found for exactly one — the assignment of [6], kept by [7, 8, 9]. None of those papers remarks on the choice.

The status of this observation is weaker than the theorem’s, and is stated as such. The group sweep is exhaustive and proves that no *image* of a known code is admissible under any of the 16 colourings, so admissible sets have to be built by repair — and repair is a search: fixed-cardinality local search under each colouring for 1500 s reached $o = 322$ under the published colouring and nothing admissible under the other fifteen. For those fifteen the honest statement is that 1500 s of search found nothing, not that nothing exists. The control that makes the negative reading meaningful: started from the published auxiliary set under the published colouring, the same search returns $o = 321$, exactly the value of [7].

Observation 9 (the count is a property of the code, not of its size). An independent rebuild of the Polak–Schrijver construction, agreeing with the printed set in 365 of its 367 words, admits only 6 candidate private pairs; the linear 343-word code admits none at all.

Two words out of 367 change the count by two: nothing about the size, or about the circular-graph construction, forces eight. That is the reason to look for a different count outside this construction, and not to expect one within it.

6. Reproducibility

The computations above are exact integer computations, and the artefacts are published with the note.

The verifier. Independence is checked by a standalone C program with no dependencies. For $n \geq 4$ two distinct vertices of $C_n^{\boxtimes d}$ are adjacent exactly when the boxes $\prod_i \{x_i, x_i + 1\}$ meet, so a set is independent iff its $|I| \cdot 2^d$ box cells are all distinct; the verifier sweeps them into an occupancy bitmap in

one linear pass, in integer arithmetic. It reports the size, the dimension, the verdict and the SHA-256 of the input file, and it knows nothing about how a set was produced.

What the verifier is held to. A verifier that always answered INDEPENDENT would pass every reproduction, so the calibration is made of cases where it can fail: 1177 single-vertex corruptions against independently computed ground truth (including 18 that leave an independent set, which it must accept); maximality by exhaustive addition over all 7^d vertices; the box method against a plain quadratic pair test at $d \leq 4$; the chunked low-memory path against the single-pass path; and negative controls. A last test asks whether those tests have teeth: a deliberately defective build of the same verifier, seeing only repeated vertices, calls a dependent set independent and is rejected.

The sets. Every set is published as one vertex per line with its SHA-256, in `sets/SHA256SUMS`: the printed code (9bdb8b9d69d9451e), our rebuild of it (23826340a496cb04), the eight pipeline codes, the best repaired auxiliary set (a4009726c06beb61) and the linear 343-word code (fc4dcf8e7e214c8f).

Seconds against hours. Theorem 6 and Proposition 5 are a matter of seconds: verifying a 367-word set sweeps 11,744 box cells, and the enumeration is one pass over 16807 vertices. (As a cross-check, the 134753-word set of [6] in $C_7^{\boxtimes 10}$ verifies in 0.47 s, 137,987,072 cells.) What needs a full run is the group sweep (663.9 s) and the sixteen auxiliary searches (1500 s each); the whole study used about 18.7 core-hours and no GPU time.

How the computations were produced. Through Claude Code (Claude Opus 5) under the author’s direction, with the safeguards part of the process: predictions registered before each run, every estimator calibrated on a case where it could fail (the defective verifier above among them), and every number in this note reconciled by script against the computation behind it, an unsourced number counting as a failed build.

Independence from the code. Theorem 6 does not rest on the software: its proof is Lemma 3 plus two finite checks, each verifiable from the printed set in [5] in a few lines of any language. The programs reproduce those lists; they are not a premise of the argument.

7. What remains open

1. *A 367-word code from outside the Polak–Schrijver construction.* The count is a property of the individual code (Observation 9), and within the construction the counts run $5, \dots, 8$. A code with nine candidates would move the record by $1.7 \cdot 10^{-4}$; producing one needs a new construction, or a direct attack on $\alpha(C_7^{\boxtimes 5})$, where even 368 is open.
2. *A base gadget in a dimension other than five.* Every published gadget lives in $C_7^{\boxtimes 5}$ because that is where the 367 lives; nothing in the product lemma requires it.
3. *The recursion, not the base.* The heterogeneous framework of [9] extracts $2.725 \cdot 10^{-5}$ more than the homogeneous one from the same base gadget — more than the entire o -route on this code could give — so any future base improvement is worth more through it.

Data and code availability. The sets, the scripts that produce every number in this note, the machine reconciliation of those numbers against the stored results, and the literature check are archived at doi:10.5281/zenodo.22979509 [10].

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